Machine Intelligence 1 Tutorial

11 Inference & Bayesian networks

Fachgebiet Neuronale Informationsverarbeitung (NI)

WiSe 25/26

Inference and Uncertainty
- The probabilistic model
- Evidence
  - Marginalization ("summing out")
  - Applying the product rule

Independence
- Motivation
- Bayes' theorem
- Using Bayes' theorem
- The Naive Bayes model

Bayesian Networks (part 1/2)
- Directed Acyclic Graphs (DAG)
  - A "Californian" example
- Preparations to perform inference
  - Topological sorting

References

"As far as the laws of mathematics refer to reality, they are not certain; and as far as they are certain, they do not refer to reality."

Albert Einstein, 1921

Inference and Uncertainty

Because we need to

quantify our degree of belief/

quantify our uncertainty

The probabilistic model

input	
has_fur	1
barks	0
meows	1

$\longrightarrow$ model $\longrightarrow$ P(cat=true)

0.99

The probabilistic model

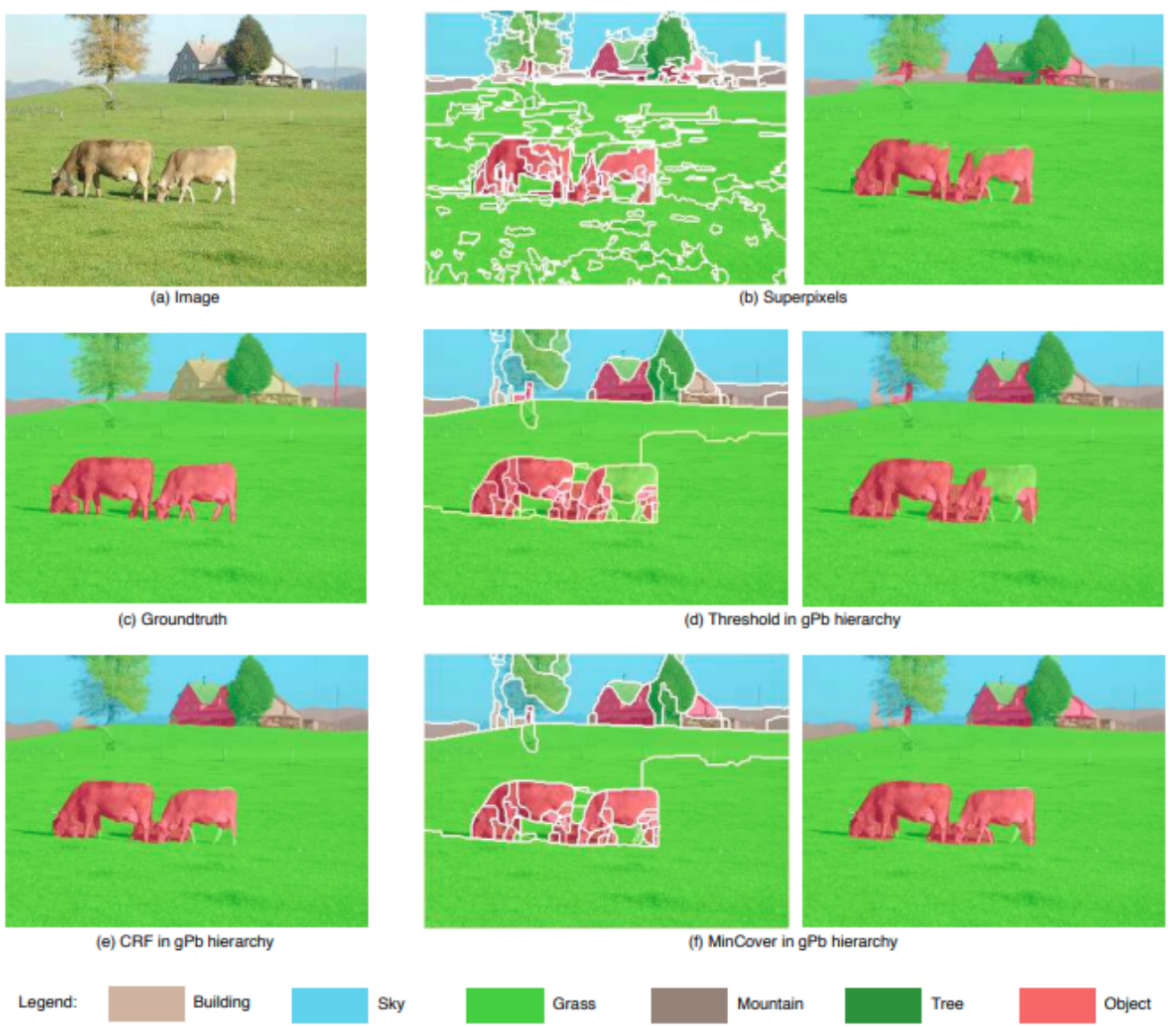

Figure: Semantic segmentation (pixel-level classification) (Farabet et al., 2012)

The probabilistic model

- <u>the model:</u>

$$
\begin{aligned}
P(H) : H \to [0,1] & \\
P(H) = 0 \quad & \iff H \text{ is false (i.e. impossible)} \\
P(H) = 1 \quad & \iff H \text{ is true (i.e. certain)} \\
0 < P(H) < 1 \quad & \text{quantifies our degree of belief}
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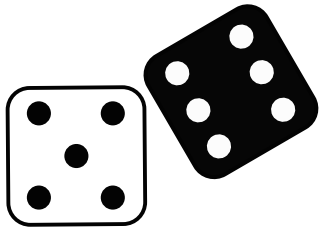
- <u>events</u>: assignment to a set of cases (e.g. two dice add up to 11)



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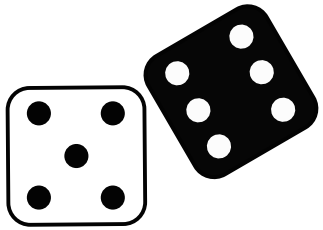

- events: assignment to a set of cases (e.g. two dice add up to 11)
- proposition: disjunction of **events** (i.e. isolate the sets of **events** in which the **proposition** holds)
  - e.g. $cavity =$ true is equivalent to:
    $(cavity = \text{true} \wedge toothache = \text{false} \wedge catch = \text{false}) \vee$
    $(cavity = \text{true} \wedge toothache = \text{true} \wedge catch = \text{true}) \vee \ldots$

$\underbrace{\textit{Summing over}}_{\text{marginalization}}$ the probabilities of the **events**

$\rightsquigarrow$ probability of **proposition**.

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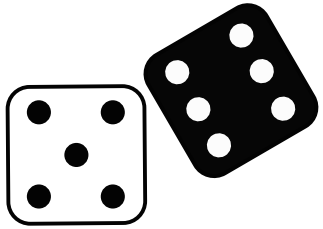
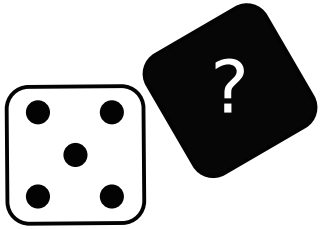
Example with dice:



$$\overbrace{P(total = 11)}^{\text{A}} = \overbrace{P(die_1 = 5, die_2 = 6)}^{\text{B}} + \overbrace{P(die_1 = 6, die_2 = 5)}^{\text{C}}$$
$$= 1/36 + 1/36$$
$$= 1/18$$

The terms A,B and C are *unconditional* or *prior* probabilities.
They represent the degree of belief in the absence of any other information.
Priors are also referred to as *domain knowledge*.

Evidence



What if we already *know* (i.e. observed) something?
e.g. We've already rolled the first die and got 5.
Now we're waiting to roll the second die.
We are now interested in the *conditional* (or *posterior*) probability of the second die leading to a total of 11:

$$P(\ total = 11 \underbrace{\ \mid\ }_{\text{“given”}} \overbrace{die_1}^{\{1,2,3,4,5,6\}} = 5\ )$$

A thing about priors

The prior is valid even after the outcome.

e.g.

$$P(\overbrace{cavity}^{\text{shorthand for } cavity=\text{true}}) = 0.2$$

$$P(cavity|\overbrace{toothache}^{toothache=\text{true}}) = 0.6$$

If we observe that a $cavity$ has actually occurred, we don't have to change our prior on cavities.

However, the prior becomes less useful if we proceed to infer other things.

Interpreting conditional probabilities

$$P(cavity|toothache) = 0.6$$

Caution

- × Whenever $toothache$ is true, then $cavity$ is true with probability of 0.6
- ✓ Whenever $toothache$ is true, <u>and no other information</u> is available, then $cavity$ is true with probability of 0.6.

other information: e.g. diagnosis was <u>no</u> $cavity$. In this case, the probability is no longer 0.6, but:

$$P(cavity \mid toothache \wedge diagnosis_{cavity} = \text{false}) \overset{\text{e.g.}}{=} 0.02$$

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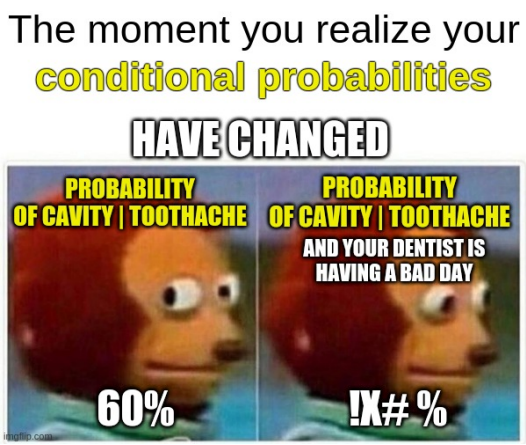

We were talking about evidence

Observing event e rules out events where e is false. This leaves a set where $P(e) > 0$.

Therefore, when we query for some variable x, we look within that set for the fraction that satisfies x and e:

$$P(x|e) = \frac{P(x, e)}{P(e)}$$

which brings us to the ***product rule***:

$$P(x, e) = P(x|e)P(e)$$

Q: *What information do I need to calculate the probability of any proposition?*

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- The full joint probability distribution
Example with $Cavity, Toothache$ and $Catch$:

	$toothache =$ true		$toothache =$ false	
	$catch =$ true	$catch =$ false	$catch =$ true	$catch =$ false
$cavity =$ true	0.108	0.012	0.072	0.008
$cavity =$ false	0.016	0.064	0.144	0.576

Marginalization ("summing out")

$$P(x|\underline{\mathbf{e}}) = \frac{P(x, \underline{\mathbf{e}})}{P(\underline{\mathbf{e}})} = \alpha P(x, \underline{\mathbf{e}}) = \alpha \sum_{\underline{\mathbf{y}} \in \underline{\mathbf{Y}}} P(\underline{\mathbf{x}}, \underline{\mathbf{e}}, \underline{\mathbf{y}})$$

where

x: query variable

$\underline{\mathbf{e}}$: evidence variable(s)

$\underline{\mathbf{y}}$: **un**observed variables in the table

α: normalization factor:

Q: *How do we calculate the normalization factor α?*

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1. evidence prior:

$$\frac{1}{\alpha} := P(\underline{\mathbf{e}}) = \sum_{x, \underline{\mathbf{y}}} P(x, \underline{\mathbf{e}}, \underline{\mathbf{y}})$$

2. or simply whatever ensures $\sum_x P(x, \underline{\mathbf{e}}) \stackrel{!}{=} 1$

Applying the product rule

	$toothache = \text{true}$		$toothache = \text{false}$	
	$catch = \text{true}$	$catch = \text{false}$	$catch = \text{true}$	$catch = \text{false}$
$cavity = \text{true}$	0.108	0.012	0.072	0.008
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$$
\begin{aligned}
P(cavity|toothache = \text{true}) &= \frac{P(cavity, toothache)}{P(toothache)} \\
&= \frac{0.108 + 0.012}{0.108 + 0.012 + 0.016 + 0.064} \\
&= 0.6
\end{aligned}
$$

Applying the product rule

	$toothache = \text{true}$		$toothache = \text{false}$	
	$catch = \text{true}$	$catch = \text{false}$	$catch = \text{true}$	$catch = \text{false}$
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$$P(cavity|toothache = \text{true}) = 0.6$$

$$P(cavity = \text{false}|toothache = \text{true}) = ?$$

Q: *What should we expect?*

Applying the product rule

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Q: *What should we expect?*

$$P(cavity = \text{false}|toothache = \text{true}) = \frac{P(cavity = \text{false}, toothache)}{P(toothache)}$$

$$= \frac{0.016 + 0.0064}{0.108 + 0.012 + 0.016 + 0.064}$$

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&= \frac{0.016 + 0.0064}{0.108 + 0.012 + 0.016 + 0.064} \\
&= 0.4 = 1 - 0.6
\end{aligned}
$$

Remark: The denominator remains constant no matter which value of $cavity$ we are interested in.
$\Rightarrow$ use $1/P(toothache)$ as the normalization factor for $P(Cavity|toothache)$.

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$$
\begin{aligned}
P(\overbrace{Cavity}^{\begin{pmatrix} cavity = \text{true} \\ cavity = \text{false} \end{pmatrix}} | toothache) &= \alpha P(Cavity, toothache) \\
&= \alpha \big[P(Cavity, toothache, catch) \\
&\qquad + P(Cavity, toothache, catch = \text{false}) \big] \\
&= \alpha \left[\begin{pmatrix} 0.108 \\ 0.016 \end{pmatrix} + \begin{pmatrix} 0.012 \\ 0.064 \end{pmatrix} \right] \\
&= \alpha \begin{pmatrix} 0.12 \\ 0.08 \end{pmatrix} \overset{\alpha = 1/P(toothache)}{=} \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}
\end{aligned}
$$

OR normalize s.t. $\sum_{\underline{\mathbf{x}}} P(x, \underline{\mathbf{e}}) \overset{!}{=} 1$

$$
= \alpha \begin{pmatrix} 0.12 \\ 0.08 \end{pmatrix} \overset{\alpha = \frac{1}{0.12 + 0.08}}{=} \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix}
$$

Independence

Statistical independence: $P(X, Y) = P(X)P(Y)$

Independence: Motivation

The full joint distribution has everything we need to perform inference, but it scales badly with more and more variables (e.g. adding $Weather$ (e.g. # of weather conditions = 4) leads to a table with $2 \times 2 \times 2 \times 4 = 32$ entries)

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Q: *Should the weather have any influence on cavities,* $toothache$ *or the dentist?*

$$P(toothache, catch, cavity, Weather)$$
$$\stackrel{!}{=} P(toothache, catch, cavity)P(Weather)$$

The 32 elements in the table can be split into 8 + 4 .

Independence: Motivation

"Extreme" case:
n independent coin flips: 2^n combinations.
Independence allows us to reduce this to $n\times$ single-variable distributions.

Bayes' theorem

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

How do we get there? - product rule
Why is it important? $\Rightarrow$ Conditional independence

Bayes' theorem

Start with the product rule:

$$P(X|Y)P(Y) = P(Y, X) = P(X, Y)$$

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$$P(X|Y)P(Y) = P(Y|X)P(X)$$

solve for $P(Y|X)$ to get Bayes' rule

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

How to read Bayes' theorem

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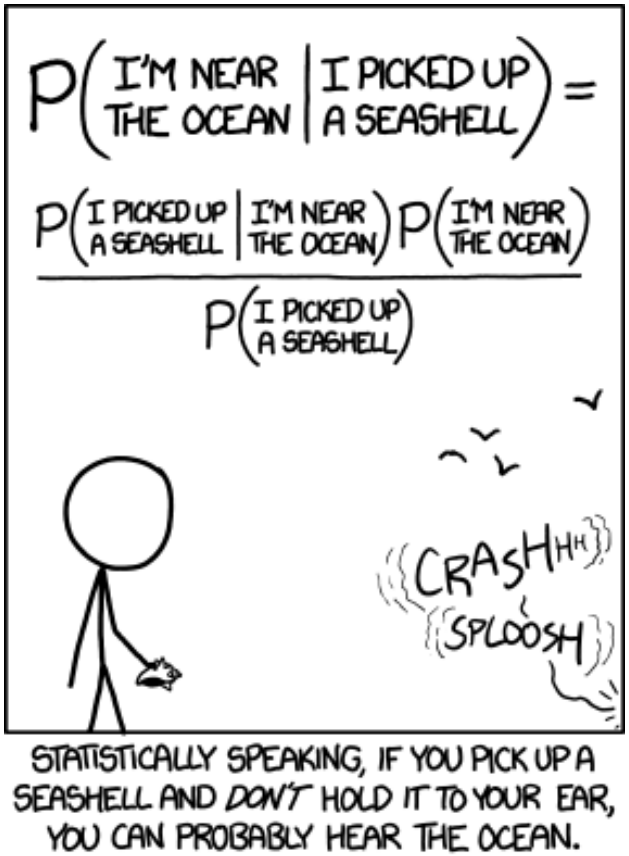

Figure: Bayes' Theorem example from xkcd.com

How to read Bayes' theorem

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

$$\underbrace{P(cause|effect)}_{\text{diagnositc}} = \frac{\overbrace{P(effect|cause)}^{\text{causal}}\,P(cause)}{P(effect)}$$

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$$P(Y|X) = \alpha P(X|Y)P(Y)$$

α can be chosen to whatever will make the entries in $P(Y|X)$ sum up to 1.

Using Bayes' theorem

$$P(Y|X) = \alpha P(X|Y)P(Y)$$

$$\begin{aligned} &P(Cavity|toothache = \text{true}, catch = \text{true}) \\ &\quad = \alpha \underbrace{P(toothache = \text{true}, catch = \text{true}|Cavity)}_{\circledast} P(Cavity) \end{aligned}$$

$\circledast$ similar scaling problem, not much better than the case of using the full joint distribution.

Using Bayes' theorem + independence?

$$
\begin{aligned}
&P(Cavity|toothache, catch) \\
&\quad = \alpha P(toothache, catch|Cavity)P(Cavity)
\end{aligned}
$$

Q: *Can we use independence to further mitigate the scaling problem here?*

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$Toothache$ given $cavity$ is independent of $Catch$ given $cavity$.

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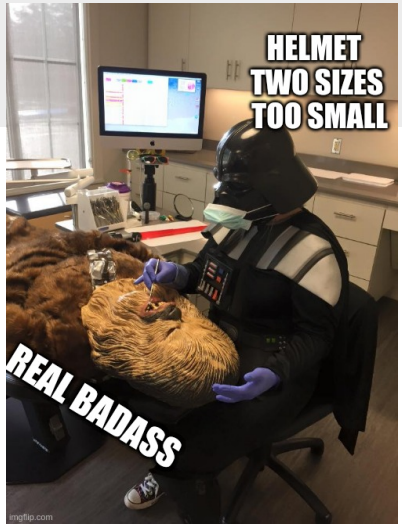
- $toothache$ depends on $cavity$: nerves & tolerance for pain.
- $catch$ depends on $cavity$: dependent on how skillful the dentist is.
- **But**...

Using Bayes' theorem + independence



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- **But**. . . "nerves & tolerance for pain" is <u>independent</u> of "how skillful the dentist is".

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$$= \alpha P(toothache, catch|Cavity)P(Cavity)$$

"nerves & tolerance for pain" is <u>independent</u> of "how skillful the dentist is". Therefore:

$$P(toothache, catch|Cavity)$$
$$= P(toothache|Cavity)P(catch|Cavity)$$

i.e. the *conditional* independence of $toothache$ and $catch$ <u>given</u> $Cavity$

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The Naive Bayes model

An alternate expression for the full joint distribution by exploiting conditional independence.
We start with the product rule:

$$P(X, Y) = P(X|Y)P(Y)$$

$$P(Toothache, Catch, Cavity) = P(Toothache, Catch|Cavity)\underbrace{P(Cavity)}_{\text{prior}}$$

Exploiting conditional independence yields:

$$\begin{aligned} &P(Toothache, Catch, Cavity) \\ &\quad = P(Toothache|Cavity)P(Catch|Cavity)\underbrace{P(Cavity)}_{\text{prior}} \end{aligned}$$

The Naive Bayes model

$$P(Toothache, Catch, Cavity)$$
$$= P(Toothache|Cavity)P(Catch|Cavity)\underbrace{P(Cavity)}_{\text{prior}}$$

In terms of cause and effect:

$$P(Cause, Effect_1, Effect_2, \ldots, Effect_n)$$
$$= \underbrace{P(Cause)}_{\text{prior}} \prod_{i=1}^{n} P(Effect_i|Cause) \qquad (1)$$

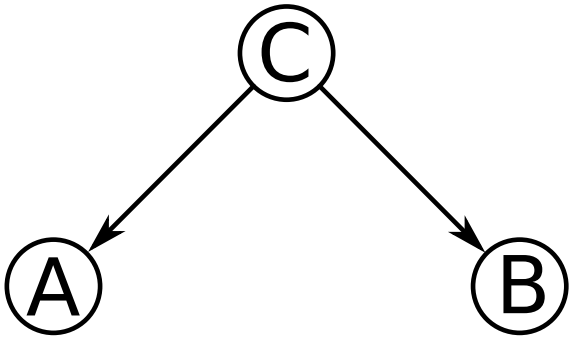
Complexity reduced!
(1) is the formulation of the *Naive Bayes model*.

Bayesian Networks (part 1/2)

Towards an efficient representation of the full joint distribution that exploits conditional independence

Directed Acyclic Graphs (DAG)

A graphical representation of conditional independence:



This leads to the factorization of the full joint distribution:

$$P(A, B, C) = P(A|C)P(B|C)P(C)$$

Directed Acyclic Graphs (DAG): toothache example

Q: *What does the DAG representation of the full joint distribution look like for toothache example?*

$$P(Toothache, Catch, Cavity)$$
$$= P(Toothache|Cavity)P(Catch|Cavity)P(Cavity)$$

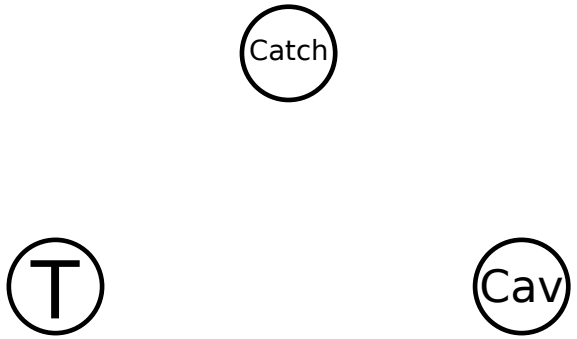


Figure: DAG representation for the toothache example.

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$$P(Toothache, Catch, Cavity)$$
$$= P(Toothache|Cavity)P(Catch|Cavity)P(Cavity)$$

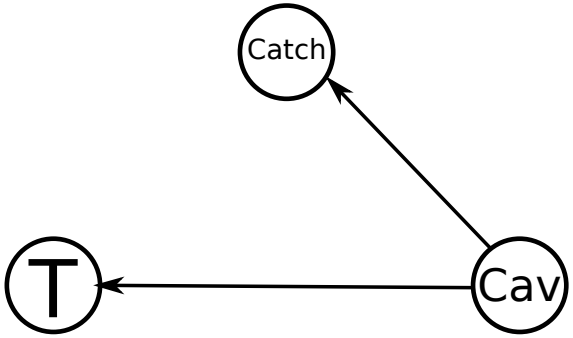
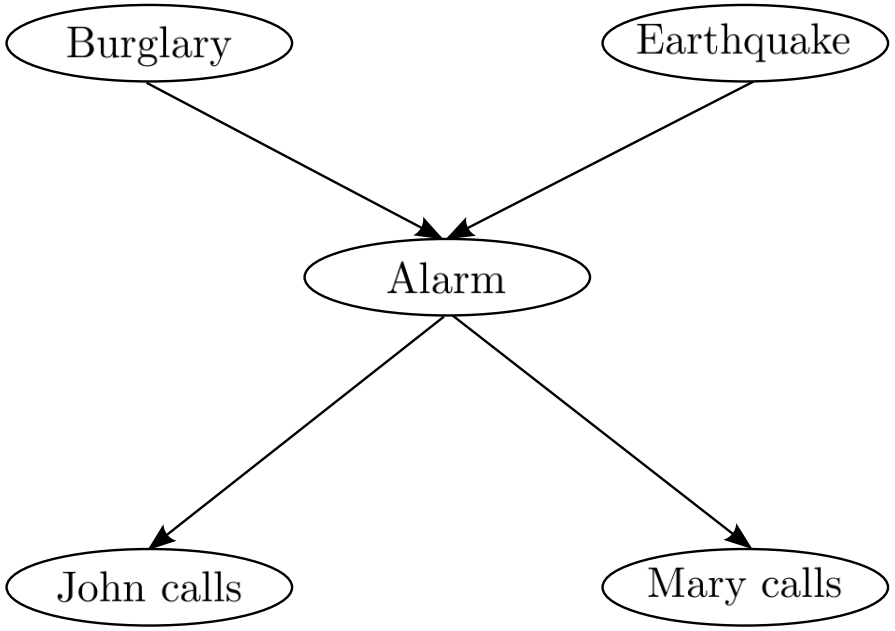
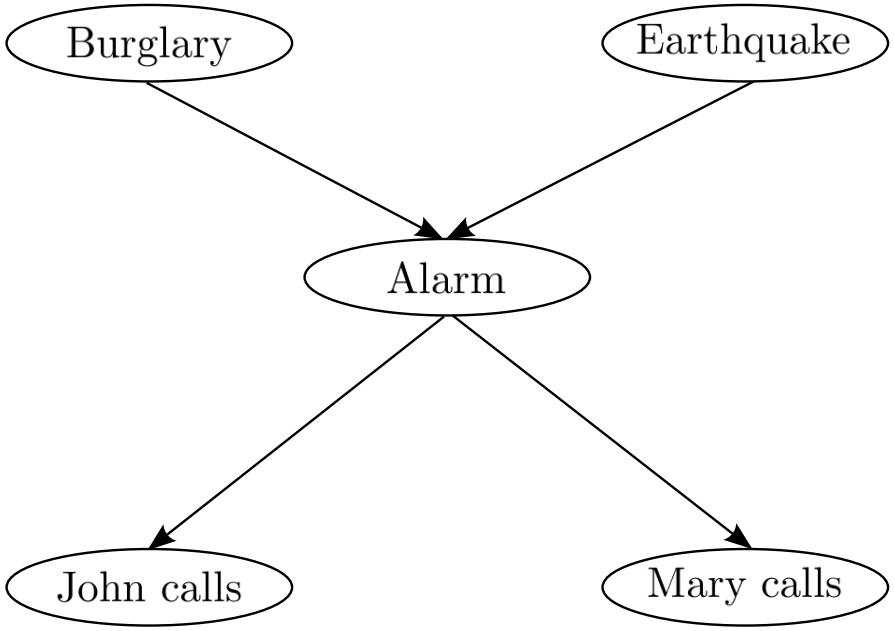


Figure: DAG representation for the toothache example.

A "Californian" example

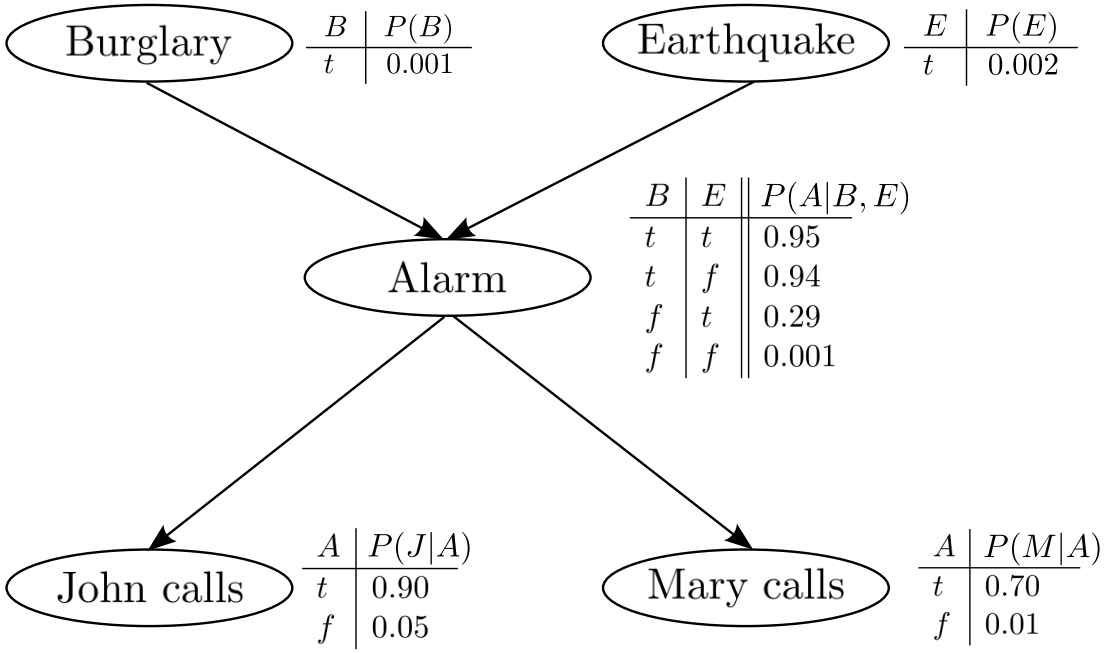


A "Californian" example



$$P(J, M, A, B, E) \overset{\text{product rule}}{=} P(J|M, A, B, E)\, P(M|A, B, E)\, P(A|B, E)\, P(B|E)\, P(E)$$

A "Californian" example



B	$P(B)$
t	0.001

E	$P(E)$
t	0.002

B	E	$P(A\|B,E)$
t	t	0.95
t	f	0.94
f	t	0.29
f	f	0.001

A	$P(J\|A)$
t	0.90
f	0.05

A	$P(M\|A)$
t	0.70
f	0.01

Considering statistical dependencies, only 10 instead of 31 entries have to be stored.

$$P(J,M,A,B,E) \overset{\text{product rule}}{=} P(J|M,A,B,E)\,P(M|A,B,E)\,P(A|B,E)\,P(B|E)\,P(E)$$

$$\overset{\text{cond. indep.}}{=} P(J|A)\,P(M|A)\,P(A|B,E)\,P(B)\,P(E)$$

Bayesian Networks (part 1/2)

A factorization of the full joint distribution[1]:

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i | parents(X_i))$$

Remember: independence is important. Independence can be identified via the Markov blanket

The Markov blanket of a node includes:

- its parents
- its children
- other parents of the children

[1] See Ch. 14 from (Russell and Norvig, 2016) for more details.

Preparations to perform inference

1. use topological representation to ensure a compact representation.
   The purpose of topological sorting is to identify cliques in the chordal decomposable graph (next week)
2. turn DAG into a *moral graph* i.e.
   an undirected graph + new edges s.t. each node of the original DAG is now directly connected to the nodes of its *Markov blanket*
3. Perform inference efficiently (next week)
   by applying *message passing* on the *Junction tree*

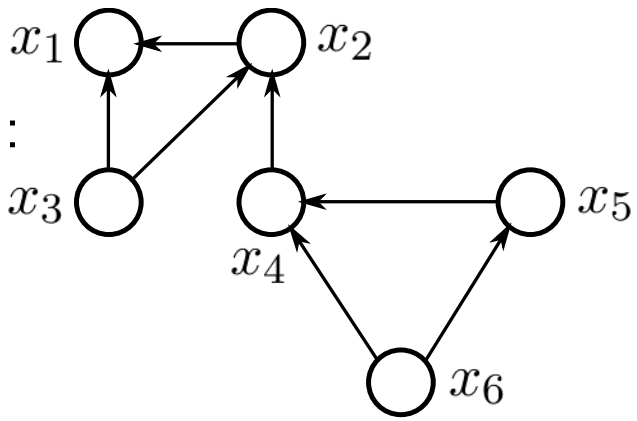
Topological sorting

- Factorization of the unconditional probability:

$$P(X_1, \ldots, X_n) \quad = \quad \prod_{i=1}^{n} P\big(X_i \,|\, parents(X_i)\big)$$

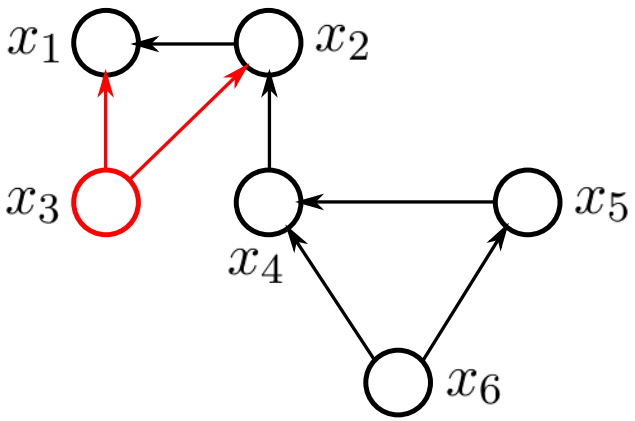
- Edges should always be directed from nodes with lower to nodes with higher indices.

- Topological sorting: From parents to children:
  1. select and queue a node without parents
  2. delete that node from the DAG
  3. repeat until DAG is empty
  4. renumber nodes



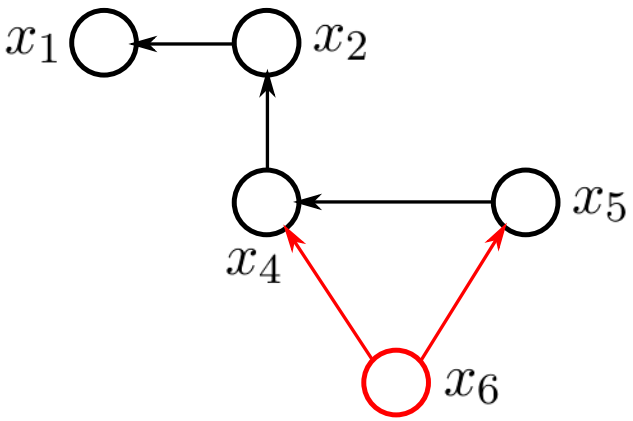
Topological sorting

- example:
  - x_3,



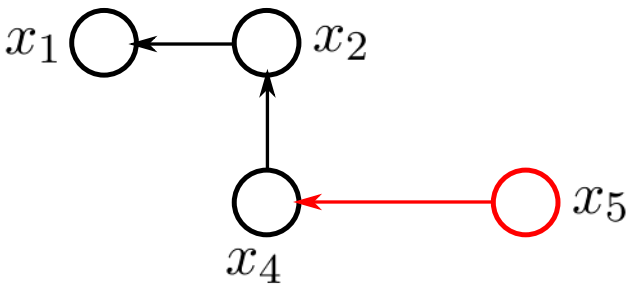
Topological sorting

- example:
  - $x_3, x_6,$



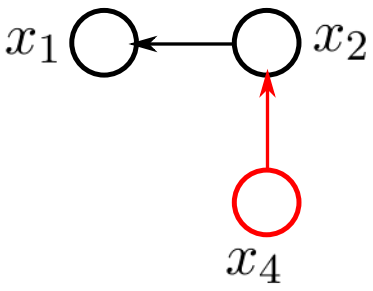
Topological sorting

- example:
  - $x_3, x_6, x_5,$



Topological sorting

- example:
  - $x_3, x_6, x_5, x_4,$



Topological sorting

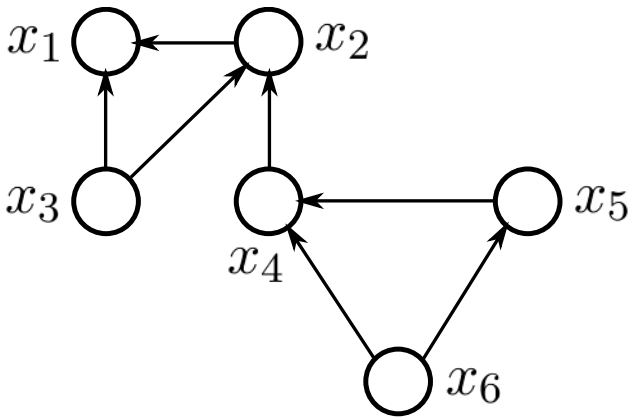
- example:
  - $x_3, x_6, x_5, x_4, x_2,$

Topological sorting

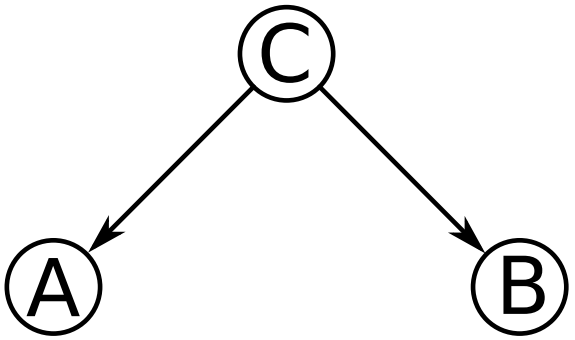
- example:
  - $x_3, x_6, x_5, x_4, x_2, x_1$

Topological sorting

- example:
  - $x_3, x_6, x_5, x_4, x_2, x_1$
- other possible topological orderings:
  - $x_6, x_5, x_4, x_3, x_2, x_1$
  - $x_6, x_5, x_3, x_4, x_2, x_1$
  - $x_6, x_3, x_5, x_4, x_2, x_1$



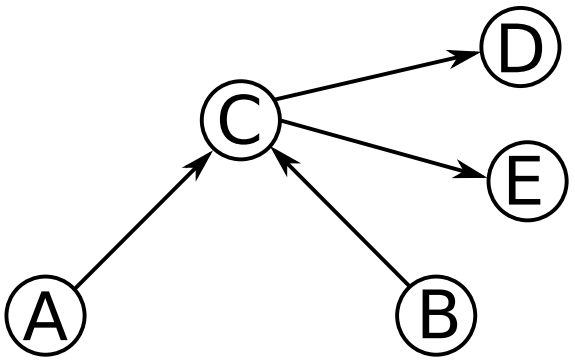
Exercise: topological sorting



- Topological sorting: From parents to children:
  1. select and queue a node without parents
  2. delete that node from the DAG
  3. repeat until DAG is empty
  4. renumber nodes

(see blackboard)

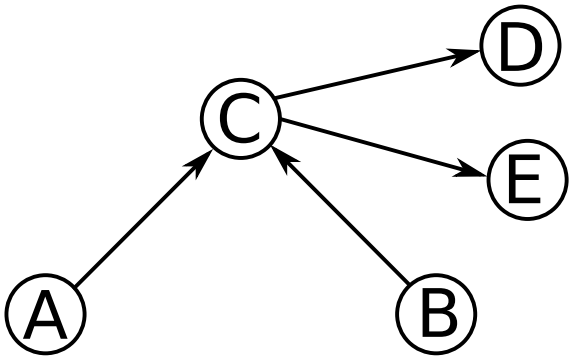
Exercise: topological sorting



- Topological sorting: From parents to children:
  1. select and queue a node without parents
  2. delete that node from the DAG
  3. repeat until DAG is empty
  4. renumber nodes

(see blackboard)

Exercise: DAG (same as before) to moral graph



1. identify the Markov blanket of each node (parents, children, other parents of the children)
2. make the edges undirected
3. add edges such that all nodes are directly connected to the Markov blanket.

(see blackboard)

References I

Clement Farabet, Camille Couprie, Laurent Najman, and Yann LeCun. Learning hierarchical features for scene labeling. *IEEE transactions on pattern analysis and machine intelligence*, 35(8):1915–1929, 2012.

Stuart J Russell and Peter Norvig. *Artificial intelligence: a modern approach*. Malaysia; Pearson Education Limited,, 2016.